Simulation and Modeling Lab(7B1)

Q1.

A university researcher wants to determine whether the average marks of students taught using two different teaching methods are significantly different.

- Method A: [32+d, 35+d, 30+d, 34+d, 31+d, 36+d, 33+d, 29+d]
- Method B: [38+d, 40+d, 36+d, 42+d, 39+d, 37+d, 41+d, 43+d]

Here, d = the last two digits of your student ID.

Write a Python program to:

a) Calculate the mean and sample standard deviation of each group.

b) Perform an independent two-sample t-test assuming equal population variances.

c) Display the t-statistic and p-value.

d) At a 5% significance level, determine whether there is a significant difference between the two methods.

Q2.

A university expects students to prefer four programming languages equally. A survey of 100 students produces the following observed frequencies:

- Python: 35
- C: 25
- Java: 20
- C++: 20

Write a Python program to:

a) Store the observed frequencies in a NumPy array.

b) Calculate the expected frequency for each category .

c) Perform a chi-square goodness-of-fit test.

d) Display the chi-square statistic and p-value.

e) At a 5% significance level, determine whether students' preferences differ significantly from an equal distribution.